

A Synopsis

Of

Project

On

AI-Powered Continuous Student Learning Analytics and Teacher Feedback System

Submitted in partial fulfilment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE & ENGINEERING
(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)**

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SEPTEMBER 2026

CERTIFICATE

This is to certify that the project synopsis entitled “AI-Powered Continuous Student Learning Analytics and Teacher Feedback System” is being submitted by Akhilesh Padiyar, Ayush Bhandari and Vaishnavi Gupta for the partial fulfillment of the requirement for the award of the degree of B. Tech- CSE (Artificial Intelligence and Machine Learning) in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, and is a record of their own work carried out under my supervision. The matter embodied in the synopsis has not been submitted in any other university or institution for the award of any degree.

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DECLARATION

We hereby declare that the project synopsis entitled “AI-Powered Continuous Student Learning Analytics and Teacher Feedback System” is being submitted for the partial fulfillment of the requirement for the award of the degree of B. Tech- CSE (Artificial Intelligence and Machine Learning) in Department of Computer Science & Engineering, Graphic Era Deemed to be University, Dehradun, is our original work and has not been submitted to any other university or institution for the award of any degree or diploma.

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ABSTRACT

Modern educational environments generate large amounts of student assessment data, but teachers often receive only overall marks or simple subject-level averages. Such information may show that students are underperforming without clearly identifying the particular topics or concepts responsible for the difficulty. A continuous assessment system can provide more useful information by analysing student responses at topic and subtopic level and converting the results into understandable learning insights.

The proposed system is an AI-powered student-teacher feedback platform that aims to reduce the effort required to continuously assess learning and interpret classroom performance. The system will maintain subjects and topics covered in class, generate periodic assessments using Generative AI, evaluate student responses, and calculate topic-level performance. Individual results will be used to identify strengths and weaknesses, while aggregated classroom results will reveal topics in which a significant number of students are struggling.

The proposed system will also maintain historical assessment and learning-performance information. Previous results can be analysed to identify improvement or persistent weaknesses over time. Retrieval-Augmented Generation (RAG) can be used with teacher-provided learning material to support targeted explanations and recommendations. An LLM will receive structured performance evidence and generate clear, actionable feedback for students and teachers. A web dashboard will present topic mastery, strengths, weaknesses, class-level learning gaps, performance trends, and recommended teaching interventions.

The initial implementation will use a sample academic environment containing subjects, syllabus topics, teacher-provided study material, question banks, student responses, and periodic assessments. The project therefore provides a practical academic implementation of Generative AI, natural language processing, embeddings, RAG, learning analytics, recommendation systems, backend development, and data processing for continuous educational feedback.

1. Introduction

1.1 Background

Modern educational systems commonly use examinations and periodic tests to measure student performance. However, a final score or subject average does not always reveal which concepts a student understands or where learning gaps exist. Teachers may therefore need to manually inspect answers and compare performance across many students before deciding which topics require additional attention.

For example, a student may score 70% in a DBMS test while performing strongly in SQL but poorly in normalization. A class average may also hide the fact that a large proportion of students are struggling with the same topic. Determining these learning gaps requires analysing individual responses, topic mappings, difficulty levels, and aggregated classroom performance rather than relying only on overall marks.

1.2 Continuous Student Assessment and Feedback

Continuous student assessment involves measuring what students have learned, identifying concepts that remain difficult, and providing feedback that can guide future learning and teaching. Traditional assessment platforms generally focus on marks, ranks, or completion status. They may not automatically convert assessment responses into topic-level insights or provide teachers with a clear picture of collective classroom learning gaps.

When the number of students, subjects, topics, and assessments increases, manual analysis becomes time-consuming. An intelligent feedback system can reduce this effort by combining assessment data, topic mappings, historical performance, and learning material and presenting the evidence in a structured form for both students and teachers.

1.3 Topic-Level Learning Analytics

Topic-level learning analytics is concerned with identifying the concepts that most strongly contribute to a student's or class's learning difficulty. A low overall score does not necessarily

mean that all topics are weak. Therefore, question-to-topic mapping, accuracy, difficulty, response patterns, and previous performance can be considered together when identifying strengths, weaknesses, and priority learning gaps.

1.4 AI-ASSISTED EDUCATIONAL FEEDBACK

Generative AI can assist in creating assessment questions from teacher-provided topics and learning material, while natural language processing and semantic embeddings can support retrieval of relevant educational content. Historical performance can provide evidence about persistent or improving weaknesses. Retrieval-Augmented Generation can supply relevant material to a language model so that performance findings can be converted into understandable and actionable feedback.

1.5 MOTIVATION

The motivation behind the proposed system is to bring assessment generation, performance analysis, learning-gap detection, personalized feedback, and teacher recommendations into a single continuous workflow. Instead of stopping at marks or grades, the system aims to move from assessment toward evidence-based topic analysis, historical comparison, targeted intervention, and reassessment. This demonstrates how Generative AI, NLP, RAG, learning analytics, and recommendation techniques can be applied to a practical educational problem.

2. Literature Review

2.1 Continuous Assessment and Learning Analytics

Continuous assessment and learning analytics have become important because educational environments generate performance information across many students, subjects, assessments, and learning activities. Earlier approaches focused mainly on test scores, grades, and statistical summaries, while modern learning analytics seeks to identify patterns in student behaviour and performance.

Research in learning analytics has demonstrated that assessment results can be used to identify patterns in student performance and support instructional decisions. Such approaches show the importance of analysing multiple indicators rather than relying on a single final score. This requirement becomes more important when a teacher manages many students across multiple topics and assessments.

Learning analytics therefore needs to consider assessment responses, topic mappings, difficulty levels, historical results, and classroom-level distributions together. The proposed system follows this direction by collecting structured assessment data and using it as the basic evidence for student and teacher feedback.

2.2 Topic-Level Performance Analysis

Root cause analysis attempts to distinguish the original cause of an incident from the symptoms that appear later in dependent components. Research on distributed-system debugging has shown that identifying the responsible component is difficult when failures propagate across service boundaries. Techniques based on execution paths, temporal ordering, dependency information, and performance relationships can help narrow the set of possible causes.

In a practical classroom, an overall test score is not sufficient to identify the learning gap, so multiple signals need to be considered together. This becomes especially important when students have mixed performance across topics. The proposed system considers question accuracy, topic-level scores, difficulty, class distributions, and historical trends while ranking learning priorities.

The objective is to provide evidence-based insights that can support a teacher's instructional decisions.

2.3 Assessment Analytics and Learning-Gap Detection

Assessment records contain detailed information about questions, topics, difficulty, correctness, and student responses. These records can be analysed to identify unusual or consistently weak performance patterns. Statistical measures can identify topics whose mastery differs substantially from the class baseline, while machine-learning techniques can be explored for more complex performance patterns.

The proposed system combines numerical performance indicators with semantic information from learning material and feedback. This allows the platform to use both quantitative assessment evidence and textual educational context rather than relying only on marks.

2.4 Student Performance Correlation and Trend Analysis

Relationships between questions, topics, subtopics, difficulty levels, and assessments provide important information about how performance should be interpreted. If several questions mapped to the same concept are answered incorrectly, the pattern provides stronger evidence of a learning gap than one isolated incorrect answer. Correlation and trend analysis can further help determine whether performance changes are related to particular topics or interventions.

In the proposed system, the academic environment is represented using relationships among subjects, topics, subtopics, questions, and assessments. The system uses these mappings to understand where incorrect responses are concentrated. The resulting evidence is combined with topic scores, difficulty, and historical performance so that isolated mistakes are less likely to be treated as major learning gaps.

2.5 Machine Learning for Student Performance Analysis

Machine-learning-based student performance analysis can be useful because fixed pass/fail thresholds may not capture changes across different assessments and topics. Statistical methods

such as mean, standard deviation, percentile analysis, and Interquartile Range can identify unusually weak or strong performance, while machine-learning methods can be explored to identify more complex student-performance patterns. For a project prototype, combining interpretable statistical measures with machine-learning techniques provides a practical basis for comparison.

The proposed system will therefore consider topic mastery thresholds, mean and standard deviation, percentile analysis, IQR-based analysis, and selected machine-learning techniques for identifying performance patterns. Detected learning gaps will be passed to later stages rather than being treated as final diagnoses. This separation allows performance analysis and AI recommendation generation to perform different roles within the overall system.

2.6 Historical Performance Retrieval and Similarity Analysis

Previous assessment results can provide useful context when a new performance pattern resembles an earlier learning difficulty. A historical performance repository can contain subjects, topics, assessment scores, common errors, interventions, and subsequent outcomes. Simple keyword matching may be insufficient when educational concepts or learning material are expressed using different terms.

Semantic embeddings provide a way to represent learning content and feedback as numerical vectors so that relevant material can be retrieved based on meaning. The proposed system can use embeddings to compare current learning gaps with stored educational content and previous feedback. This information can help identify recurring weaknesses and provide additional context for recommendations. Historical retrieval is therefore treated as a supporting stage rather than a replacement for current assessment analysis.

2.7 Retrieval-Augmented Generation for Educational Feedback

Retrieval-Augmented Generation combines information retrieval with language-model generation. Instead of asking a language model to produce feedback only from its internal knowledge, relevant information is first retrieved from teacher-provided notes, syllabus material, approved resources, or historical learning records and supplied as context. This approach is useful because educational content and teaching strategies are specific to the subject and classroom.

In the proposed system, the identified learning gap and relevant topic information can be used to retrieve appropriate learning resources. The most relevant material can then be provided as supporting context to the language model along with structured performance evidence. This design helps keep the generated feedback grounded in available educational content and reduces unsupported recommendations.

2.8 Large Language Models for Assessment and Feedback

Large language models provide new possibilities for converting assessment evidence into natural-language feedback. In an educational system, an LLM can explain difficult concepts, summarize performance, suggest revision strategies, and help teachers interpret classroom trends. However, using an LLM as the sole assessment or diagnosis component may introduce unsupported conclusions. The proposed system therefore places the LLM after the main analytical stages.

Assessment evaluation, topic-level scoring, classroom aggregation, trend analysis, and relevant-content retrieval first produce structured evidence. The LLM then receives this evidence and generates concise student feedback or teacher recommendations. This separation keeps the core performance analysis dependent on measurable assessment information while using the language model mainly for explanation and actionable communication.

2.9 Research Gap

Existing educational technologies address online testing, learning analytics, automated question generation, recommendation, and AI tutoring as separate or partially connected capabilities. A practical educational prototype that connects GenAI-based assessment generation, topic-level performance analysis, student feedback, classroom learning-gap detection, historical analysis, RAG, and LLM-based recommendations in one workflow can provide a useful integrated perspective.

The proposed system addresses this gap by combining these stages around a simulated academic environment. The project does not attempt to replace teachers or make high-stakes educational decisions automatically. Instead, it focuses on demonstrating an evidence-driven feedback pipeline in which each stage contributes information to the next stage.

3. PROBLEM STATEMENT & OBJECTIVES

3.1 Problem Statement

Modern distributed applications can produce large volumes of performance metrics and application logs. When a failure occurs, several services may show abnormal behaviour within a short period because of their dependencies. A monitoring platform may therefore generate multiple alerts for the same underlying incident. The main difficulty is to determine which event represents the actual root cause and which events are only secondary effects.

The proposed problem is to develop an intelligent system that can generate periodic assessments from topics covered in class, analyse student responses at topic and subtopic level, identify individual strengths and weaknesses, aggregate classroom performance, retrieve relevant learning material, and generate actionable feedback using AI. The system should also maintain historical performance so that improvement and persistent learning gaps can be identified. The proposed workflow therefore combines GenAI-based assessment generation, automated evaluation, learning analytics, historical comparison, RAG, and AI-based feedback.

3.2 OBJECTIVES

The main objectives of the project are:

1. **To identify unexpected drops in student performance** and learning progress across continuous evaluations.
2. **To group related learning difficulties** across quizzes and topics into a structured student profile.
3. **To pinpoint specific topics and basic concepts** where students face maximum difficulty.
4. **To link test scores, question difficulty, time taken, and learning topics** to find underlying patterns in student learning.
5. **To map out how different topics build on top of fundamental prerequisite concepts** within the course syllabus.

6. **To review student answers and written feedback** to understand common misconceptions and mistakes.
7. **To find past student cases with similar learning challenges** and study patterns to offer tested remedies.
8. **To bring in relevant textbook chapters and lecture notes** to support personalized study plans and guidance.
9. **To generate clear, simple, and easy-to-read feedback and guidance** for teachers and students.
10. **To provide an easy-to-use web dashboard** showing performance summaries, weak topics, underlying reasons for gaps, past examples, and clear improvement recommendations.

4. PROPOSED METHODOLOGY



Fig. 1. End-to-end operational workflow of the proposed system.

4.1 OVERALL SYSTEM ARCHITECTURE

The proposed AI-Powered Continuous Learning Analytics and Teacher Feedback System is structured as a modular incident investigation platform designed to ingest metrics and logs, execute anomaly detection, build dependency structures, perform root cause analysis (RCA), and

generate human-readable explanations via Retrieval-Augmented Generation (RAG). The overall architecture separates data collection, processing, inference, and visualization to maintain flexibility across distributed microservice environments.

The target environment consists of a web application used by teachers and students. The teacher defines subjects, topics, syllabus coverage, and approved learning resources. Controlled assessment cycles are then created at suitable intervals, such as every two weeks. The system generates questions, evaluates responses, stores performance records, and produces student- and classroom-level insights. The architecture is designed so that the assessment cycle can be repeated throughout the semester.

4.2 Dataset Collection

Data collection gathers syllabus information, covered topics, teacher-approved learning material, generated questions, question metadata, student responses, assessment scores, and historical performance. Each question is associated with a topic and difficulty level so that responses can later be aggregated into meaningful learning indicators. The system also stores teacher interventions and subsequent assessment results where applicable.

4.3 Data Preprocessing

Raw educational data requires normalization before downstream analysis. Assessment records are validated for missing or inconsistent values, question-topic mappings are checked, and scores are standardized where required. Learning material passes through text extraction and chunking pipelines so that meaningful sections can be represented as embeddings. Student identifiers are separated from analytical records wherever possible to support privacy-aware processing.

4.4 ASSESSMENT GENERATION AND VALIDATION

The system evaluates metric time-series and log event frequencies using statistical and machine-learning-based anomaly detection models. Metric anomalies are flagged when performance indicators deviate beyond adaptive statistical thresholds computed over rolling historical windows.

Log stream anomalies are identified by tracking abrupt deviations in event log distribution rates using baseline models (e.g., isolation forests or autoencoders), marking period spikes in error-level logs as candidate anomalies

4.5 STUDENT ASSESSMENT AND EVALUATION

Students attempt the generated assessment through the student interface. Responses are automatically evaluated for objective question types, while structured response information is stored with the corresponding question, topic, difficulty, and assessment identifier. Scores are then calculated at both assessment and topic level.

Temporal correlation algorithms cluster concurrent metric spikes and error log bursts into a unified incident event, establishing a single timeline context for deep-dive investigation.

4.6 TOPIC AND QUESTION MAPPING

the proposed AI-Powered Continuous Student Learning Analytics and Teacher Feedback System maintains a dynamic topological map representing service relationships and communication pathways.

Students, subjects, topics, questions, assessments, and resources are represented as connected entities. A question is associated with a topic, an assessment contains multiple questions, and a student's response is linked to the corresponding question and topic.

Topic and question mappings can be updated whenever the teacher adds new syllabus content, modifies covered topics, or approves new questions. This keeps the analytical structure aligned with the current course content.

4.7 STUDENT LEARNING-GAP ANALYSIS

To identify learning gaps, the analytics module correlates student responses with topic mappings, difficulty levels, score distributions, and previous assessment performance. The system distinguishes isolated incorrect answers from repeated weak performance across questions belonging to the same concept.

Topic-level analysis calculates indicators such as accuracy, mastery percentage, question coverage, and performance change over time. Weak topics are prioritized according to the proportion of students below a defined mastery threshold and the consistency of poor performance across related questions.

4.8 CLASSROOM AGGREGATION AND TREND ANALYSIS

Learning-gap candidates are assigned priority scores using factors such as topic mastery, proportion of students struggling, question difficulty, repeated errors, and change from previous assessments. The resulting ranking provides a prioritized list of topics that may require revision or additional practice.

4.9 LEARNING MATERIAL RETRIEVAL AND NLP

Once potential learning gaps are identified, relevant teacher-provided notes, syllabus sections, and learning resources associated with those topics are passed to an NLP and retrieval pipeline.

Learning resources and feedback content are converted into vector representations using sentence-transformer or other suitable embedding models. This enables semantic retrieval of relevant explanations and examples even when the wording of the learning gap differs from the wording in the source material.

4.10 RETRIEVAL-AUGMENTED GENERATION (RAG)

The context collected during an active learning analysis—consisting of topic mastery, common errors, assessment history, relevant topic information, and retrieved teacher-approved learning resources—is combined into a structured prompt context. This RAG framework grounds the downstream generation phase on evidence relevant to the student's or class's identified learning needs.

4.11 LLM-BASED FEEDBACK AND RECOMMENDATIONS

The structured RAG prompt is passed to a Large Language Model (LLM) to produce coherent, human-readable feedback. Rather than presenting students and teachers with raw scores alone, the LLM synthesizes structured evidence to explain weak concepts, suggest revision strategies,

recommend remedial activities, and generate targeted practice questions. Teacher feedback is based on aggregated classroom evidence and approved learning context.

4.12 STUDENT AND TEACHER DASHBOARD

The output from the learning analytics pipeline is visualized through web-based student and teacher dashboards. Key elements include subject-wise performance, topic mastery, strength and weakness indicators, class-level learning-gap charts, historical trends, recommended interventions, relevant learning resources, and AI-generated feedback.

5. HARDWARE AND SOFTWARE REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

1. **Processor:** Intel Core i5 or equivalent processor
2. **RAM:** Minimum 8 GB, with 16 GB recommended
3. **GPU:** NVIDIA GPU with CUDA support recommended for model fine-tuning and comparison
4. **GPU Memory:** 4 GB or higher recommended
5. **Storage:** Minimum 20 GB free space
6. **System:** Desktop computer or laptop
7. **Internet:** Stable internet connection for software packages, APIs, and development resources

5.2 SOFTWARE REQUIREMENTS

1. **Operating System:** Windows / Linux
2. **Programming Language:** Python
3. **Data Processing:** Pandas and NumPy
4. **Anomaly Detection:** Isolation Forest and statistical methods.
5. **Database:** PostgreSQL.
6. **NLP and Embeddings:** Sentence Transformers
7. **Vector Search:** FAISS or Chroma.
8. **RAG:** LangChain or a custom retrieval pipeline.
9. **Machine Learning:** Scikit-learn
10. **LLM:** Suitable API-based large language model.
11. **Backend:** FastAPI.
12. **Dashboard:** Streamlit.
13. **Containerization:** Docker.
14. **Development Environment:** VS Code / Jupyter Notebook
15. **Version Control:** Git and GitHub
16. **Browser:** Google Chrome / Microsoft Edge or any modern web browser

6. EXPECTED OUTCOMES

The proposed system is expected to produce the following results:

1. **Detection of abnormal learning behaviour** and performance drop-offs from continuous evaluation metrics.
2. **Creation of structured learning gap profiles** from related topic-level anomalies.
3. **Identification of affected subtopics**, core concepts, and their prerequisite dependencies.
4. **Ranking of possible root causes** of learning gaps using temporal performance trends, difficulty correlation, subtopic dependencies, and error severity.
5. **Retrieval of similar historical learning gaps** and remedial cases using embeddings and vector search.
6. **Use of RAG (Retrieval-Augmented Generation)** to provide grounded course syllabus and lecture context to the learning gap analysis.
7. **Generation of a human-readable, actionable pedagogical feedback** and study plans using an LLM.
8. **A web dashboard** showing student/class learning gap details, supporting evidence, root-cause mastery scores, similar historical performance cases, and generated feedback explanations.
9. **A prototype architecture** that can later be extended toward real-time classroom monitoring and university-scale educational analytics.

7. CONCLUSION

The proposed system is an AI-powered continuous student learning analytics and teacher feedback platform. The project addresses the difficulty of identifying specific learning gaps when traditional assessments provide only overall scores. Instead of limiting the system to test generation and marks, the proposed approach combines assessment data, topic-level analytics, historical performance, learning-resource retrieval, and Generative AI to provide useful feedback.

The proposed methodology begins with collecting syllabus information, teacher-approved learning material, and assessment data. Generative AI is used to create questions aligned with covered topics, followed by automated evaluation and topic-level performance analysis. Classroom aggregation, historical trends, and semantic retrieval are then used to identify priority learning gaps and relevant educational resources. This creates a structured evidence base for generating feedback.

The final stage uses Retrieval-Augmented Generation and a large language model to convert structured performance evidence into clear student feedback and actionable teacher recommendations. The dashboard provides a consolidated view of topic mastery, strengths, weaknesses, classroom learning gaps, trends, and suggested interventions. Overall, the project demonstrates a practical combination of Generative AI, NLP, embeddings, RAG, LLMs, learning analytics, recommendation systems, backend development, and data processing for a real-world educational problem. The prototype can serve as a foundation for future work involving adaptive learning, personalized practice, predictive learning-gap detection, and controlled AI-assisted interventions.

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